

LEGO - Lego

#scc (/problems/tag/scc)

It's Christmas morning, and you've got what you wanted: a box of Lego™! (Okay, maybe not, but better than nothing.)

Lego is pretty fun to tinker with, and you've decided to build some sort of shape. (For the sake of this problem, let's say your shape is basically 2-dimensional - it'll be a slab.)

But once you pick it up, you discover that you didn't plan it properly, and your wonderful shape just falls apart. Now, you're planning to build something big, and so you're going to use the computer to help you.

Write a program, that given the layout of a Lego design, outputs the number of pieces it would break into if picked up.

- The Legos will be built on a x-y coordinate plane, with (0, 0) being the bottom left corner.
- The blocks are flat on your carpet, so a block will never 'fall down'.
- If you haven't seen a Lego brick before: A Lego brick has grooves on its top that match with notches on the bottom. If a groove and a notch bind, the bricks will stay together. See the diagram.
- Assume that the bricks bind together perfectly. A brick will bind with another brick securely even if just a single notch touches another groove.

Input

The first line contains N (the number of Lego pieces), $1 \leq N \leq 100000$.

N lines follow, each with 4 integers x_1, y_1, x_2, y_2 ($0 \leq x_1 < x_2 \leq 2 \times 10^9$, $0 \leq y_1 < y_2 \leq 2 \times 10^9$)

This means that there is a brick with bottom left corner (x_1, y_1) and top right corner (x_2, y_2) . x denotes the horizontal coordinate and y the vertical coordinate. Two bricks will bind if one's bottom y -coordinate coincides with the other's top y -coordinate and the intersection of the two intervals (the bottom of one and the top of the other) has non-zero length.

No bricks will overlap.

Output

A single line containing the number of separate pieces that these blocks form.

Example

Input:

4

0 0 2 2

1 2 3 4

2 0 4 2

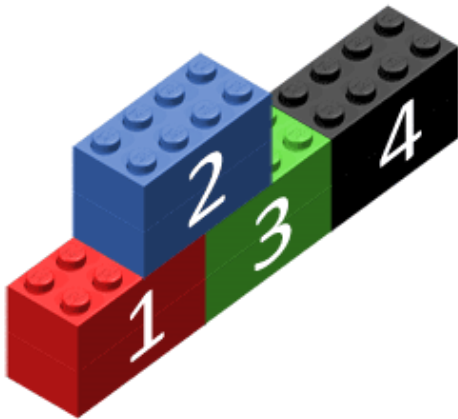
4 0 6 2

Output:

2

Explanation

Blocks #1, 2, 3 are joined securely. However, #4 is just hanging around.



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llite_27 (/users/llite_27): 2022-09-14 14:17:58

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karelisp (/users/karelisp): 2021-08-04 23:20:00

I get the error "Missing testcases: 'NoneT'" upon submission. What does it mean ?



wslord (/users/wslord): 2021-04-27 21:50:21

The question has some problem in submission. It says "Missing testcases: 'NoneT'" on submitting any solution



kanishk779 (/users/kanishk779): 2019-03-24 14:29:15

How should we construct the graph?



luka_dzimba911 (/users/luka_dzimba911): 2019-03-15 16:12:00

Dont think too much, try to think of simple solution :D
And dont use SPOJ TOOLKIT testcases to debug they have some wrong testcases

Last edit: 2019-03-15 16:13:03



Sajal Sarkar (/users/cnatka): 2016-12-28 11:02:23

Did someone solve this question via counting connected component?

Last edit: 2016-12-28 11:02:51



SANYAM JAIN (/users/jsanyam): 2016-08-03 19:01:48

Finally solved it, took so many wa and tle.



lalitesh (/users/lalitesh): 2016-08-02 20:55:23

After WA n TLE , got AC ;)



Shubham Sinha (/users/sinhashubham95): 2016-05-28 22:38:29

dsu rocks !!



birdie (/users/gudia9): 2015-02-20 10:53:27

i am getting wa.. any tricky testcase ?

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Added by: Brian Bi (/users/bbi5291)
Date: 2009-04-10
Time limit: 1s
Source limit: 50000B
Memory limit: 1536MB
Cluster: Cube (Intel G860) (/clusters/)
Languages: All
Resource: Hanson Wang
(<http://spoj.com/users/hansonw1>)

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